

OCR Further Mathematics
Pure Core - Matrices
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

1(a)

You are given that the matrix $\begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$ represents a transformation T. You are given that the line with equation $y = kx$ is invariant under T.

Determine the value of k .

[4]

(b) Determine whether the line with equation $y = kx$ in part above is a line of invariant points under T.

[1]

2(a)

The matrix A is given by $\mathbf{A} = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}$.

By considering $\mathbf{A}$, $\mathbf{A}^2$, $\mathbf{A}^3$ and $\mathbf{A}^4$ make a conjecture about the form of the matrix $\mathbf{A}^n$ in terms of n for $n \geq 1$.

[2]

(b) Use induction to prove the conjecture made in above part.

[4]

3(a) In this question you must show detailed reasoning.

The matrix A is given by $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

Define the transformation represented by A.

[1]

(b) Show that the area of any object shape is invariant under the transformation represented by A.

[1]

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The matrix B is given by $\mathbf{B} = \begin{pmatrix} 7 & 2 \\ 21 & 7 \end{pmatrix}$. You are given that B represents the transformation which is the result of applying the following three transformations in the given order.

- ♦ A shear which leaves the y -axis invariant and which transforms the point (1, 1) to the point (1, 4).
- ♦ The transformation represented by A.
- ♦ A stretch of scale factor p which leaves the x -axis invariant.

Determine the value of p .

[4]

4

Two matrices, A and B, are given by $\mathbf{A} = \begin{pmatrix} 1 & -2 & -1 \\ 2 & -3 & 1 \\ a & 1 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} -6 & 3 & -4 \\ -1 & 6 & -4 \\ 8 & -8 & -1 \end{pmatrix}$ where a is a constant.
Find the value of a for which $\mathbf{AB} = \mathbf{BA}$. [3]

5(a) A 2-D transformation T is a shear which leaves the y-axis invariant and which transforms the object point (2, 1) to the image point (2, 9). A is the matrix which represents the transformation T.

Find A. [3]



(b) By considering the determinant of A, explain why the area of a shape is invariant under T. [2]

6(a)

The matrix A is given by $A = \begin{pmatrix} 0.6 & 2.4 \\ -0.8 & 1.8 \end{pmatrix}$.

Find $\det A$.

[1]

(b) The matrix A represents a stretch parallel to one of the coordinate axes followed by a rotation about the origin.

By considering the determinants of these transformations, determine the scale factor of the stretch.

[2]

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(c) Explain whether the stretch is parallel to the x-axis or the y-axis, justifying your answer.

[1]

(d) Find the angle of rotation.

[2]

7(a)

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You are given the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}.$$

Find $\mathbf{A}^4$.

[1]

(b) Describe the transformation that A represents.

[2]

(c) The matrix B represents a reflection in the plane $x = 0$.

Write down the matrix B.

[1]

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- (d) The point P has coordinates $(2, 3, 4)$. The point P' is the image of P under the transformation represented by B .

Find the coordinates of P' .

[1]

8(a)

You are given the matrix A where $A = \begin{pmatrix} a & 2 & 0 \\ 0 & a & 2 \\ 4 & 5 & 1 \end{pmatrix}$.

Find, in terms of a , the determinant of A , simplifying your answer.

[2]

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- (b) Hence find the values of a for which A is singular.

[2]

(c) You are given the following equations which are to be solved simultaneously.

$$ax + 2y = 6$$

$$ay + 2z = 8$$

$$4x + 5y + z = 16$$

For each of the values of a found in the above part determine whether the equations have

- a unique solution, which should be found, or
- an infinite set of solutions or
- no solution.

[7]



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You are given that $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 2 \\ 3 & -2 & -1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 0 & 1 \\ -8 & 4 & 0 \\ 19 & -8 & -1 \end{pmatrix}$.

(a) Find $\mathbf{AB}$.

[1]

(b) Hence write down $\mathbf{A}^{-1}$.

[1]

(c) You are given three simultaneous equations

$$x + 2y + z = 0$$

$$2x + 5y + 2z = 1$$

$$3x - 2y - z = 4$$

(i) Explain how you can tell, without solving them, that there is a unique solution to these equations.

[2]

(ii) Find this unique solution.

[2]



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- 10 (a) Find the shortest distance between the point $(-6, 4)$ and the line $y = -0.75x + 7$. [2]

Two lines, l_1 and l_2 , are given by

$$l_1: \mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} \text{ and } l_2: \mathbf{r} = \begin{pmatrix} 11 \\ -1 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}.$$



- (b) Find the shortest distance between l_1 and l_2 . [3]

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(c) Hence determine the geometrical arrangement of l_1 and l_2 .

[2]



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The matrix A is given by $A = \begin{pmatrix} a & 2 & 3 \\ 4 & 4 & 6 \\ -2 & 2 & 9 \end{pmatrix}$ where a is a constant. It is given that if A is not singular then

$$A^{-1} = \frac{1}{24a-48} \begin{pmatrix} 24 & -12 & 0 \\ -48 & 9a+6 & 12-6a \\ 16 & -2a-4 & 4a-8 \end{pmatrix}.$$

(a) Use A^{-1} to solve the simultaneous equations below, giving your answer in terms of k .

$$\begin{aligned} x + 2y + 3z &= 6 \\ 4x + 4y + 6z &= 8 \\ -2x + 2y + 9z &= k \end{aligned}$$

[3]

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(b) Consider the equations below where a takes the value which makes A singular.

$$\begin{aligned}ax + 2y + 3z &= b \\4x + 4y + 6z &= 10 \\-2x + 2y + 9z &= -13\end{aligned}$$

b takes the value for which the equations have an infinite number of solutions.

- Determine the value of b .
- Find the solutions for y and z in terms of x .

[5]



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(a) For the equations in part (b) with the values of a and b found in part (b) describe fully the geometrical arrangement of the planes represented by the equations.

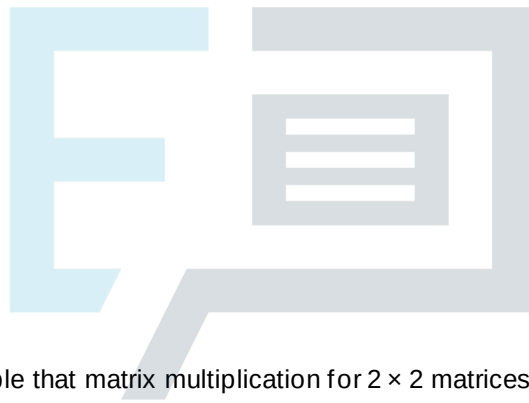
[2]

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The matrices $\mathbf{A}$ and $\mathbf{B}$ are given by $\mathbf{A} = \begin{pmatrix} 1 & a \\ 3 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 4 & 2 \\ 3 & 3 \end{pmatrix}$.

(a) Find the value of a such that $\mathbf{AB} = \mathbf{BA}$.

[3]



(b) Prove by counter example that matrix multiplication for 2×2 matrices is not commutative.

[2]

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(c) A triangle of area 4 square units is transformed by the matrix $\mathbf{B}$. Find the area of the image of the triangle following this transformation.

[1]

- (d) Find the equations of the invariant lines of the form $y = mx$ for the transformation represented by matrix B. [4]



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13 In the following set of simultaneous equations, a and b are constants.

$$3x + 2y - z = 5$$

$$2x - 4y + 7z = 60$$

$$ax + 20y - 25z = b$$

(a) In the case where $a = 10$, solve the simultaneous equations, giving your solution in terms of b .

[3]



(b) Determine the value of a for which there is no unique solution for x , y and z .

[3]

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(c) (i) Find the values of α and β for which $\alpha(2y - z) + \beta(-4y + 7z) = 20y - 25z$ for any y and z .

[3]

(ii) Hence, for the case where there is no unique solution for x , y and z , determine the value of b for which there is an infinite number of solutions.

[2]

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(iii) When a takes the value in part (b) and b takes the value in part (c)(ii) describe the geometrical arrangement of the planes represented by the three equations.

[1]

- 14 (a) Find the solution to the following simultaneous equations.

$$x + y + z = 3$$

$$2x + 4y + 5z = 9$$

$$7x + 11y + 12z = 20$$

[2]

- (b) Determine the values of p and k for which there are an infinity of solutions to the following simultaneous equations.

$$x + y + z = 3 \quad 2x + 4y + 5z = 9 \quad 7x + 11y + pz = k$$

[6]



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The matrix A is given by $A = \begin{pmatrix} a & 1 \\ 1 & 4 \end{pmatrix}$, where $a \neq \frac{1}{4}$, and I denotes the 2×2 identity matrix. Find

(i) $2A - 3I$,

[3]

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(ii) A^{-1} .

[2]



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- 16 By using the determinant of an appropriate matrix, find the values of λ for which the simultaneous equations

$$3x + 2y + 4z = 5,$$

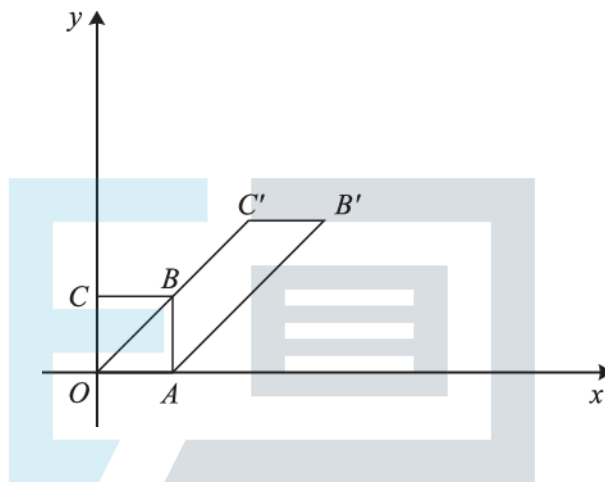
$$\lambda y + z = 1,$$

$$x + \lambda y + \lambda z = 4,$$

do not have a unique solution for x , y and z .

[6]

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The diagram shows the unit square $OABC$, and its image $OAB'C'$ after a transformation. The points have the following coordinates: $A(1, 0)$, $B(1, 1)$, $C(0, 1)$, $B'(3, 2)$ and $C'(2, 2)$.

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- (i) Write down the matrix, X , for this transformation.

[2]

- (ii) The transformation represented by X is equivalent to a transformation P followed by a transformation Q . Give geometrical descriptions of a pair of possible transformations P and Q and state the matrices that represent them.

[6]

- (iii) Find the matrix that represents transformation Q followed by transformation P .

[2]

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The matrices A, B and C are given by $A = \begin{pmatrix} a & 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} b & 0 & 5 \end{pmatrix}$ and $C = \begin{pmatrix} 6 \\ 4 \\ -1 \end{pmatrix}$. Find

(i) $5A - 3B$,

[2]



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(ii) BC,

[2]



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(iii) CA.

[2]

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The matrix $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ represents a transformation P.

(i) Describe fully the transformation P.

[2]



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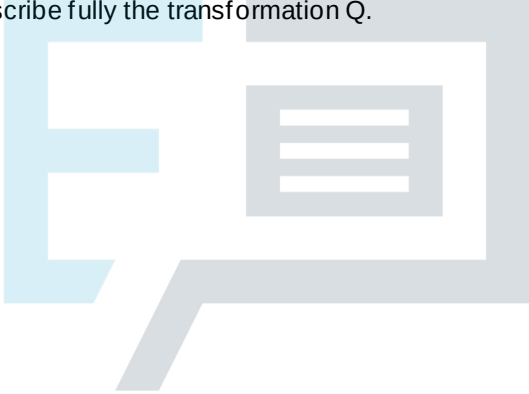
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The matrix M is given by $\mathbf{M} = \begin{pmatrix} -3 & -1 \\ -1 & 0 \end{pmatrix}$.

- (ii) Given that M represents transformation Q followed by transformation P , find the matrix that represents the transformation Q and describe fully the transformation Q .

[6]



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- (i) The matrix X is given by $\mathbf{X} = \begin{pmatrix} a & 3 & -2 \\ 0 & a & 5 \\ 1 & 2 & 1 \end{pmatrix}$. Show that the determinant of X is $a^2 - 8a + 15$.

[3]



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(ii) Explain briefly why the equations

$$3x + 3y - 2z = 1$$

$$3y + 5z = 5$$

$$x + 2y + z = 2$$

[3]

do not have a unique solution and determine whether these equations are consistent or inconsistent.



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The matrix A is given by $A = \begin{pmatrix} 2 & a \\ 0 & 1 \end{pmatrix}$, where a is a constant.

(i) Find A^{-1} .

[2]



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The matrix B is given by $\mathbf{B} = \begin{pmatrix} 2 & a \\ 4 & 1 \end{pmatrix}$.

(ii) Given that $\mathbf{PA} = \mathbf{B}$, find the matrix P.

[3]



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22

The matrix M is given by $\mathbf{M} = \begin{pmatrix} 0 & 2 \\ -1 & 0 \end{pmatrix}$.

- (i) The diagram in the Printed Answer Book shows the unit square $OABC$. The image of the unit square under the transformation represented by M is $OA'B'C'$. Draw and label $OA'B'C'$, indicating clearly the coordinates of A' , B' and C' .

[3]

- (ii) The transformation represented by M is equivalent to a transformation P followed by a transformation Q . Give geometrical descriptions of a possible pair of transformations P and Q and state the matrices that represent them.

[4]



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23

The matrix D is given by $\mathbf{D} = \begin{pmatrix} 1 & 3 & 4 \\ 2 & a & 3 \\ 0 & 1 & a \end{pmatrix}$.

(i) Find the values of a for which D is singular.

[6]



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(ii) Three simultaneous equations are shown below.

$$\begin{aligned}x + 3y + 4z &= 3 \\ 2x + ay + 3z &= 2 \\ y + az &= 0\end{aligned}$$

For each of the following values of a , determine whether or not there is a unique solution. If a unique solution does not exist, determine whether the equations are consistent or inconsistent.

(a) $a = 3$

(b) $a = 1$

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The matrices A, B and C are given by $A = \begin{pmatrix} 5 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -5 \end{pmatrix}$ and $C = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$.

(i) Find $3A - 4B$.

[2]



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(ii) Find CB. Determine whether CB is singular or non-singular, giving a reason for your answer.

[5]

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The matrix M is given by $\mathbf{M} = \begin{pmatrix} 2 & 2 \\ 0 & 1 \end{pmatrix}$. Prove by induction that, for $n \geq 1$,

$$\mathbf{M}^n = \begin{pmatrix} 2^n & 2^{n+1} - 2 \\ 0 & 1 \end{pmatrix}.$$

[6]

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(i) Find the matrix that represents a rotation through 90° clockwise about the origin.

[2]

(ii) Find the matrix that represents a reflection in the x-axis.

[2]

(iii) Hence find the matrix that represents a rotation through 90° clockwise about the origin, followed by a reflection in the x-axis.

[2]

(iv) Describe a single transformation that is represented by your answer to part (iii).

[2]



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The matrix A is given by $\mathbf{A} = \begin{pmatrix} a & 2 & 1 \\ 1 & 3 & 2 \\ 4 & 1 & 1 \end{pmatrix}$.

(i) Find the value of a for which A is singular.

[5]

(ii) Given that A is non-singular, find $\mathbf{A}^{-1}$ and hence solve the equations

$$\begin{aligned} ax + 2y + z &= 1, \\ x + 3y + 2z &= 2, \\ 4x + y + z &= 3. \end{aligned}$$

[7]

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Find the determinant of the matrix $\begin{pmatrix} a & 4 & -1 \\ 3 & a & 2 \\ a & 1 & 1 \end{pmatrix}$.

[3]

29

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The matrices A and B are given by $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ -4 & 5 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 3 & 1 \\ 2 & 3 \end{pmatrix}$ and I is the 2×2 identity matrix. Find

(i) $4\mathbf{A} - \mathbf{B} + 2\mathbf{I}$,

[2]

(ii) A^{-1} ,

[2]

(iii) $(AB^{-1})^{-1}$.

[3]



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(a)

Find the matrix that represents a shear with the y -axis invariant, the image of the point $(1, 0)$ being the point $(1, 4)$.

[2]

(b) The matrix X is given by $\mathbf{X} = \begin{pmatrix} \frac{1}{2}\sqrt{2} & \frac{1}{2}\sqrt{2} \\ -\frac{1}{2}\sqrt{2} & \frac{1}{2}\sqrt{2} \end{pmatrix}$.

(i) Describe fully the geometrical transformation represented by X .

[2]

(ii) Find the value of the determinant of X and describe briefly how this value relates to the transformation represented by X .

[2]

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